Reinforcement Learning Project Report

Snow plow problem with multi-agent reinforcement learning

Authors:

Martin Nilsson, *School of Engineering*, Jönköping University

Besan Ewir, *School of Engineering*, Jönköping University

Vladimir Giustacchini, *School of Engineering*, Jönköping University

Jorge Cuello, *School of Engineering*, Jönköping University

October 18, 2026

Contents

1 Problem Description: Snow Plow Problem **3**

2 Environment **3**
- 2.1 Rules 3
  - 2.1.1 Traffic rules 4
- 2.2 Actions 4
- 2.3 Observations 5
- 2.4 Truncation Dynamics 5
- 2.5 Rewards 6

3 Algorithms **7**
- 3.1 PPO 7
- 3.2 MAPPO 7

4 Experiments **8**
4.1 Experiment 1: 1-Agent + Full Observation 8
4.2 Experiment 2: 1-Agent + Partial Observation 8
4.3 Experiment 3: 2-Agents + Full Observation 9
4.4 Experiment 4: 2-Agents + Partial Observation 9

5 Results **10**
5.1 Experiment 1 10
5.2 Experiment 2 10
5.3 Experiment 3 10
5.4 Experiment 4 10

6 Conclusion **10**

7 Discussion **10**

1 Problem Description: Snow Plow Problem

We consider a snow plow problem where a single or group of snow plows is tasked with clearing snow from a network of roads in a city. The objective is to ~~minimize the total time taken to clear all roads while ensuring that the snow plows operate efficiently and avoid collisions~~. Each snow plow is controlled by an agent that makes decisions based on the current state of the environment, including the location of other snow plows, the amount of snow on each road, and the traffic conditions.

2 Environment

A grid represents the city, where each cell corresponds to a road segment. The state of the environment includes the positions of all snow plows, the amount of snow on each road segment, and any obstacles or traffic conditions that may affect the movement of the snow plows.

2.1 Rules

The snow plows can move in four directions (UP, DOWN, LEFT, RIGHT) and can also choose to stay in their current position (DO_NOTHING). The environment is dynamic in the sense that snow randomly falls onto road cells in the grid environment periodically based on a `cloud_prob` $= 0.02 \times$ `number_of_agents`. This is only to balanced the snow level accumulation to give the snow plows a fair difficulty across different map sizes and number of agents. and the agents must adapt their strategies accordingly.

To enforce cooperative behavior, we introduced additional rules to the environment that lets multiple agents to collaborate with snow cells with `snow_level` > 1. So agents placed in the same grid cell that faces the same snow cell will have reduced time to clear the snow compared to doing it alone and would therefore clear snow and earn rewards faster. See table 1 for more information.

`Snow_level`	**Time steps**
3	10
2	5
1	1

Table 1: Single agent plow times

`Snow_level`	**Time steps**
3	6
2	3
1	1

Table 2: Multi-agent plow times

2.1.1 Traffic rules

The agents cannot navigate the road network however they want. There are traffic rules applied for each type of grid cells that they need to follow. The environment is initialized with the intended traffic rules for each grid cell. Different rules will apply depending on the cell type and remove invalid actions that would make the agents disobey traffic rules, go into obstacle cells or attempt to go outside of the grid area for different traffic situations.

The types are as follows:

Cell type	**Color**	**Traffic rules**
ROAD	Grey	Any direction is allowed.
HIGHWAY	Light Grey	One way direction only.
INTERSECTION	Dark Grey	Remove the action that normally would lead back to going in the opposite highway direction.

Table 3: Road Network Grid Cell Types

2.2 Actions

1. $\rightarrow$: Move Right
2. $\leftarrow$: Move Left
3. $\uparrow$: Move Up
4. $\downarrow$: Move Down
5. Do nothing: Stay in the current position.

2.3 Observations

Field	Description
`grid`	-
`agent_positions`	-
`self_position`	-
`self_id`	-
`view_origin`	-

Table 4: Full and Partial Observability parameters

2.4 Truncation Dynamics

Episode ends ~~if the city streets are cleared~~ or if a maximum number of time steps is reached. The truncation dynamics ensure that the agents learn to optimize their actions within a limited time frame, promoting efficient snow clearing strategies.

- After 500 000 time steps, the episode ends.
- ~~If all roads are cleared before 1000 time steps, the episode ends early.~~

2.5 Rewards

In this snow plow environment there are some different ways the agents can earn rewards. The main rewards will of course come from clearing snow. It give +1 reward each time its clearing a snow layer, i.e. a maximum snow layer of 3 (`snow_level`= 3) will give a reward of +3.

Rewards	**Description**	**Value**
Move	Whenever a snowplow moves throughout the grid it gets a negative reward. This acts as consuming fuel.	-1
Plow snow	Snowplows gets positive rewards for clearing snow	1
Clear Main streets	An extra reward for clearing highway streets. This incentivize giving clearing priority to highways that generally in a real world scenario would probably benefit a larger portion of traffic	4
Clear secondary street	An extra reward for clearing other road streets	2

Table 5: List of rewards

3 Algorithms

3.1 PPO

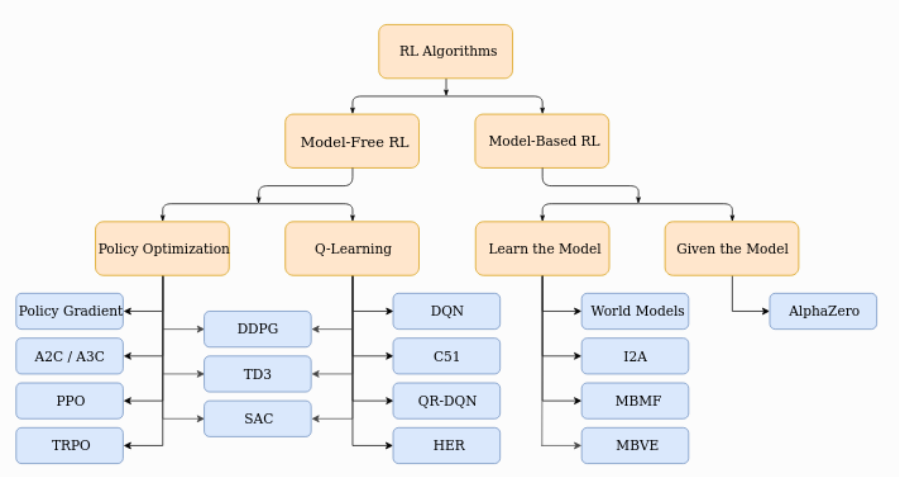


Figure 1: RL Algorithm tree

We use the **P**roximal **P**olicy **O**ptimization (PPO) algorithm that is a very popular reinforcement learning algorithm that uses policy gradient to learn optimal policies and has a built in clipping mechanism that regulates how much the action-selection strategy can change over time. This generally results in the algorithm being stable during training.

3.2 MAPPO

For experiments using multiple snow plow agents we use a Mulit-Agent version of PP0 (MAPPO) used for cooperative mulit-agent objectives. In this case, plowing snow individually or collaboratively to gain the highest collective reward.

4 Experiments

4.1 Experiment 1: 1-Agent + Full Observation

Parameter	**Value**
Number of plows	1
Observation	FULL
Algorithm	PPO
Recurrent	False
Road network	network_1

Table 6: Experiment 1 parameters

For the environment size in the first experiment we considered a grid of size 10×19 with 1 snow plow.

This experiment is meant to show the capabilities of one agent moving around the environment in an attempt to clear snow with full observability of the environment. With fill knowledge about the location of snow cells the agent should eventually after training have no problem navigating to find snow cells and

4.2 Experiment 2: 1-Agent + Partial Observation

Parameter	**Value**
Number of plows	1
Observation	FULL
Algorithm	PPO
Recurrent	False
Road network	network_1
PO-range	(3×3) or (5×5)

Table 7: Experiment 2 parameters

4.3 Experiment 3: 2-Agents + Full Observation

Parameter	**Value**
Number of plows	5
Observation	FULL
Algorithm	MAPPO
Recurrent	False
Road network	network_3

Table 8: Experiment 3 parameters

4.4 Experiment 4: 2-Agents + Partial Observation

Parameter	**Value**
Number of plows	5
Observation	FULL
Algorithm	MAPPO
Recurrent	False
Road network	network_3
PO-range	(3×3) or (5×5)

Table 9: Experiment 4 parameters

5 Results

5.1 Experiment 1

5.2 Experiment 2

5.3 Experiment 3

5.4 Experiment 4

6 Conclusion

7 Discussion

Our hopes for introducing `snow_levels` was that multi-agents would converge into a policy that tells them to collaborate on areas with more snow. But on the other hand they should also realize that it will not be optimal for all situations because working on `snow_level = 1` will still only give them +1 reward. Therefore, if they only observe level 1 cells, they would be encouraged to spread out more and cover more surface by themselves which would earn a higher total reward than doing it together all the time. This would simulate a work by demand scenario that also commonly occurs in a realistic setting.

One potential shortcoming with this environment design might be that the decision-making strategy is not strongly motivated enough by the magnitude of the time steps that it takes to clear snow in relation to the reward they are getting. For example, will one agent "give up" chasing snow cells and turn around to help another agent that "in theory" would yield a higher reward? These dynamics might not be sufficiently expressed but could potentially be improved by adding more snow levels with linear or exponential time steps needed for clearing snow. However, this would probably also increase training time.